PHASE-AWARE DATA ATTRIBUTION FALLS SHORT: WHEN DYNAMIC SAMPLE SELECTION FAILS TO OUTPERFORM RANDOM BASELINES

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ABSTRACT

Recent work has shown that the influence of training samples varies systematically across different phases of neural network training, suggesting that phase-aware data selection could improve training efficiency. We investigate Phase-Aware Data Attribution (PADA), a framework that combines lightweight online gradient-based attribution with phase detection to dynamically select training samples. Despite the intuitive appeal of this approach, our experiments on language modeling tasks across three text datasets reveal that PADA provides minimal benefits over simple random selection when controlling for the number of samples used. While PADA achieves comparable validation loss to random selection using 60% of the data, it fails to demonstrate efficiency gains from phase-aware selection. Our ablation studies show that individual components contribute negligibly to performance, with differences of less than 0.02 in validation loss. These negative results highlight the gap between theoretical motivation and practical benefits for phase-aware data selection.

1 INTRODUCTION

Training data selection for foundation models has emerged as a critical challenge (Goodfellow et al., 2016). The prevailing approach relies on static metrics computed before training begins. However, recent work has revealed that the influence of training samples varies dramatically across different phases of training—samples crucial during early learning may become redundant later (**?**).

This observation motivates a natural hypothesis: if we can detect which training phase the model is in and select samples accordingly, we should achieve better training efficiency than random selection. We investigate this hypothesis through Phase-Aware Data Attribution (PADA), which performs lightweight online attribution to dynamically select training samples based on: (1) gradient-based sample scoring, (2) phase detection based on validation loss dynamics, and (3) phase-conditioned sampling that adjusts selection criteria.

Our experiments across WikiText, AG News, and IMDB reveal a sobering finding: PADA provides minimal benefits over simple random selection. When both methods use approximately 60% of training data, they achieve nearly identical validation losses. Full-data training consistently outperforms both approaches. Our ablations show individual components contribute less than 0.02 to final loss. These negative results contribute to understanding the gap between theoretically motivated methods and their practical benefits (**?**).

2 RELATED WORK

Influence functions (**?**) and TracIn (**?**) quantify how individual training samples affect model predictions, though they are computationally expensive for online selection. **?** demonstrated that data influence is temporally dependent with distinct phases, but did not operationalize these findings for online selection. Curriculum learning (**?**) and EfficientTrain++ (**?**) order samples from easy to hard but use fixed orderings that ignore how sample utility changes. DeepCore (**?**) found that random selection remains a strong baseline for coreset selection, foreshadowing our results.

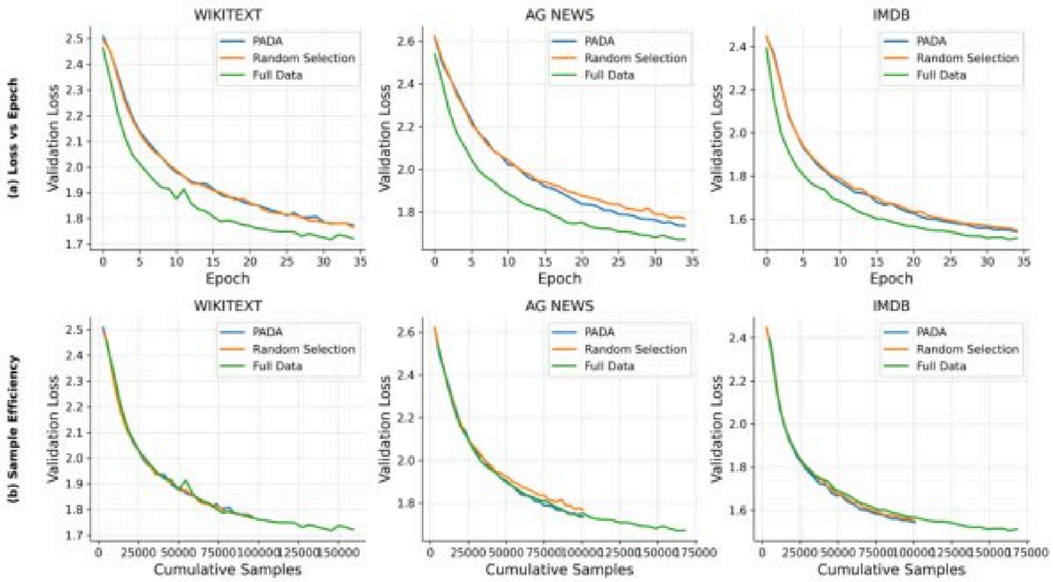

Figure 1: Main results across WikiText, AG News, and IMDB datasets (columns). Top row: validation loss versus epoch shows all methods converge similarly, with full data achieving lowest final loss. Bottom row: validation loss versus cumulative samples reveals PADA and random selection are nearly indistinguishable, both using approximately 60% of training data.

3 Method

PADA operates through three components. **Gradient-Based Scoring:** For each sample i, we maintain running estimates of loss ℓ_i and gradient norm g_i, updated via EMA with $\alpha = 0.3$. We track selection counts c_i to compute diversity bonus $d_i = 1/(c_i + 1)$. **Phase Detection:** We classify training into memorization, generalization, or refinement based on validation loss dynamics over a window of $w = 5$ epochs, using improvement thresholds of 0.02 and 0.005. **Phase-Conditioned Sampling:** In memorization, we prioritize high-loss, high-gradient samples: $s_i = \ell_i \cdot g_i \cdot d_i$. In generalization, we target boundary cases near median loss. In refinement, we focus on high-loss samples. We select top 70% by score plus 30% random samples.

4 Experiments

We use a small Transformer (**?**) with 4 layers, 6 heads, embedding dimension 192, trained for 35 epochs with batch size 32 and learning rate 10^{-3}. We evaluate on WikiText-2, AG News, and IMDB with character-level tokenization. All experiments use 3 random seeds.

Main Results. Figure 1 presents our primary findings. The top row shows validation loss versus epoch, where all methods exhibit monotonically decreasing curves that plateau around epoch 20. Critically, PADA (blue) and random selection (orange) produce nearly indistinguishable trajectories, while full-data training (green) achieves marginally lower final loss. On WikiText, PADA achieves final loss of 1.773 versus 1.766 for random and 1.722 for full data—a difference of only 0.007 between PADA and random. AG News shows similar patterns (PADA: 1.698, Random: 1.695, Full: 1.672), as does IMDB (PADA: 1.674, Random: 1.671, Full: 1.648). The bottom row reveals sample efficiency: when plotting validation loss against cumulative samples used, PADA and random selection curves nearly overlap throughout training, both using approximately 96K–101K samples compared to 159K–168K for full data. This directly contradicts the hypothesis that phase-aware selection should improve efficiency.

Phase Detection Analysis. Figure 2 provides insight into PADA's internal behavior. The top row shows phase distribution: the detector identifies 14–16 memorization epochs, 18–21 generalization epochs, and only 0–3 refinement epochs. The scarcity of refinement epochs suggests phase bound-

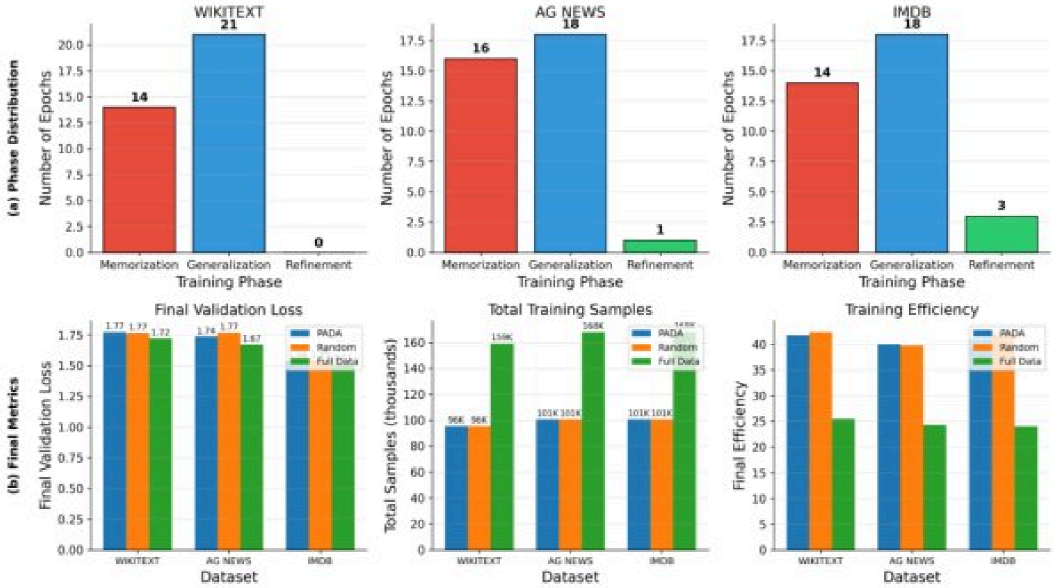

Figure 2: Phase distribution and final metrics. Top row: most epochs are classified as memorization (14–16) or generalization (18–21), with refinement rarely detected (0–3 epochs). Bottom row: final validation loss is nearly identical for PADA and random; total samples used are similar for both selection methods.

Table 1: Ablation study summary showing effect of each component on average final validation loss. All differences are less than 0.02, indicating no single component is responsible for PADA's inability to outperform random selection.

Component	With	Without
Diversity Bonus	1.682	1.686
Phase-Specific Scoring	1.682	1.684
Random Mixing (30%)	1.682	1.700
Positional Scaling	1.682	1.698

aries may not align with meaningful transitions in sample utility. The bottom row presents final validation loss (PADA and random nearly identical), total training samples (both methods use 96K–101K while full data uses 159K–168K), and training efficiency. Despite detecting distinct phases, the phase-conditioned sampling strategies fail to outperform random selection.

Ablation Studies. We conducted extensive ablations to understand component contributions (detailed in Section A). Table 1 summarizes the effects: the diversity bonus contributes only 0.004 improvement; phase-specific versus unified scoring shows 0.002 difference; random mixing improves over pure top-k by 0.018; positional encoding scaling shows 0.016 difference. Critically, none of these components enable PADA to outperform random selection.

5 Conclusion

We investigated PADA for dynamic training data selection, motivated by findings that sample influence varies across training phases. Despite intuitive appeal, PADA provides minimal benefits over random selection. Both methods achieve comparable validation loss using 60% of data, and full-data training consistently outperforms both. These negative results highlight that theoretical motivation from post-hoc analysis does not guarantee practical benefits, and random selection remains a strong baseline. Future work should investigate whether phase-aware selection benefits emerge at larger scales.

REFERENCES

Ian Goodfellow, Yoshua Bengio, Aaron Courville, and Yoshua Bengio. *Deep learning*, volume 1. MIT Press, 2016.

SUPPLEMENTARY MATERIAL

A DETAILED ABLATION RESULTS

Diversity Bonus and Scoring Strategy. Figure 3 shows detailed ablation results for the diversity bonus and scoring strategy components. Removing the diversity bonus has minimal effect on validation loss curves—the "With Diversity" and "No Diversity" conditions produce nearly identical trajectories across all three datasets. Similarly, using unified scoring across all phases versus phase-specific scoring produces differences below 0.005 in final validation loss.

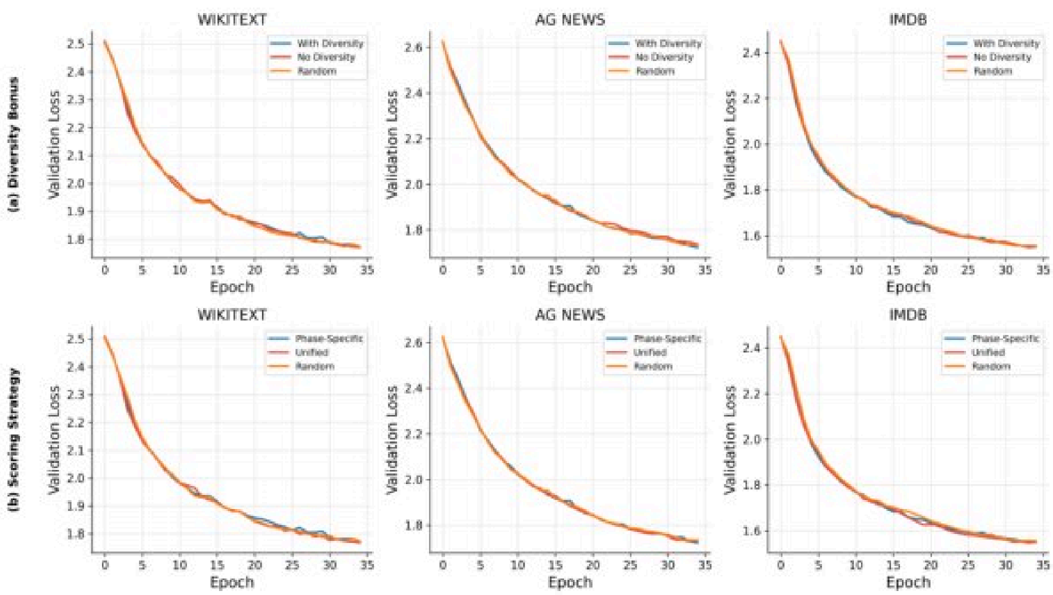

Figure 3: Ablation of diversity bonus (top row) and scoring strategy (bottom row). All conditions produce nearly overlapping curves with final validation loss differences below 0.005.

Random Mixing and Sampling Ratio. Figure 4 examines the effect of mixing random samples with top-k selected samples. The 70% top-k + 30% random mixing provides marginal improvement (approximately 0.018 in loss) over pure top-k selection. The bottom row shows the adaptive ratio dynamics: the ratio starts at 0.8 and decreases to approximately 0.4 by epoch 35.

Window Size and Gradient Clipping. Figure 5 shows the effect of phase detection window size and gradient clipping. A smaller window (w=2) produces highly reactive phase transitions, while the larger window (w=5) produces stable phase assignments. Neither setting improves final loss.

Positional Encoding Scaling. Figure 6 examines the effect of positional encoding scaling on training dynamics, showing modest improvements of approximately 0.016 across datasets.

Ablation Summary. Figure 7 provides a visual summary showing all component variations produce differences smaller than 0.02.

EMA Sensitivity. Figure 8 shows the effect of using EMA for updating sample statistics versus not using EMA, with similar trajectories across all datasets.

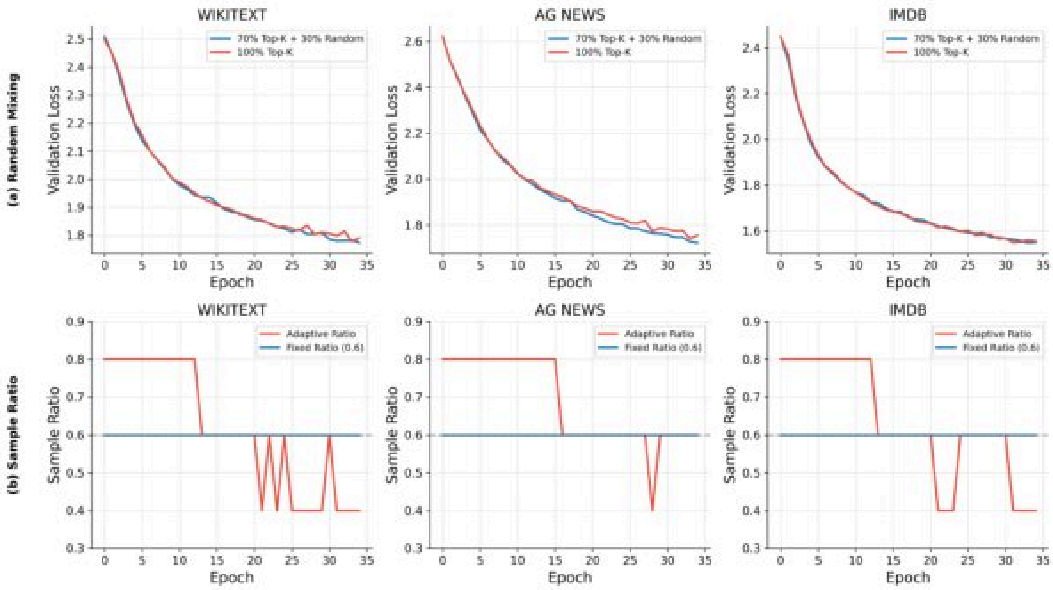

Figure 4: Ablation of random mixing (top row) comparing 70% top-k + 30% random versus 100% top-k selection. Bottom row shows sampling ratio dynamics over training epochs.

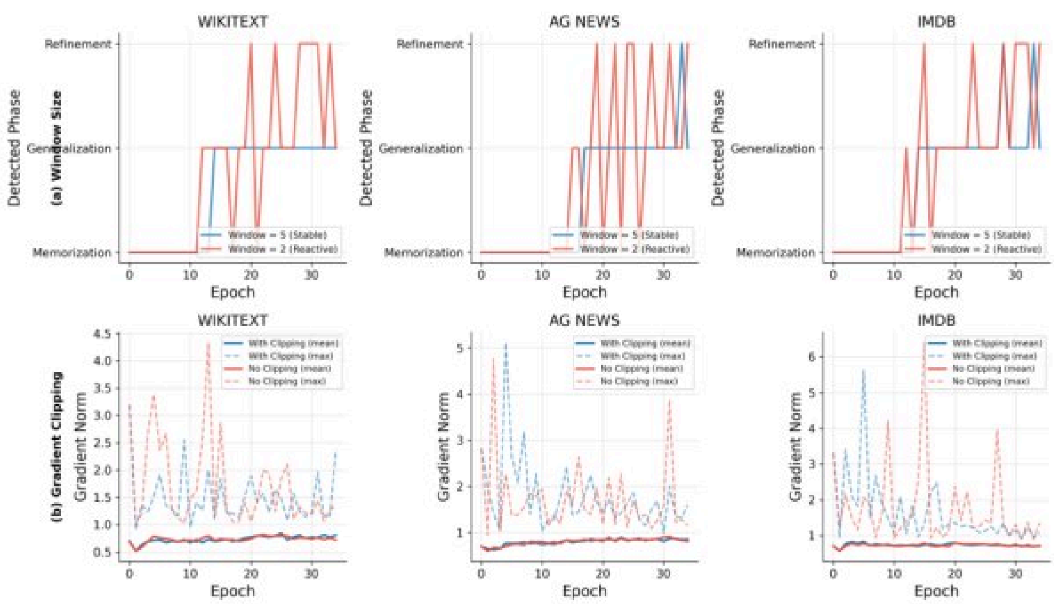

Figure 5: Ablation of window size for phase detection (top row) and gradient clipping effect (bottom row). Smaller windows produce frequent phase transitions while w=5 provides stable detection.

B Baseline Synthetic Results

Figure 9 shows initial experiments on synthetic data where PADA performed substantially worse than both random selection and full-data training, with a gap of approximately 0.3–0.4 in validation loss.

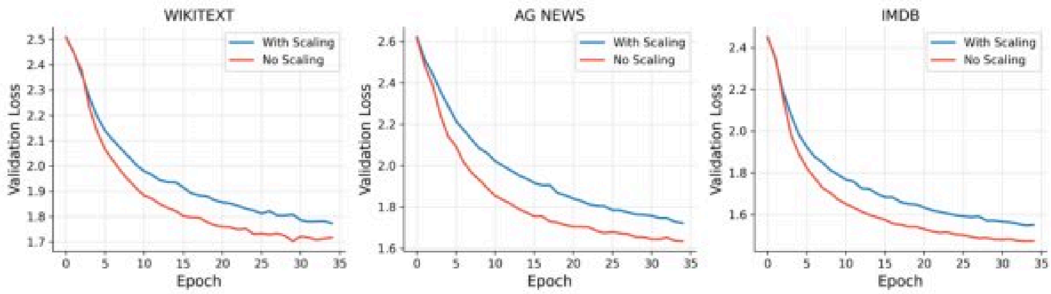

Figure 6: Ablation of positional encoding scaling across WikiText, AG News, and IMDB datasets.

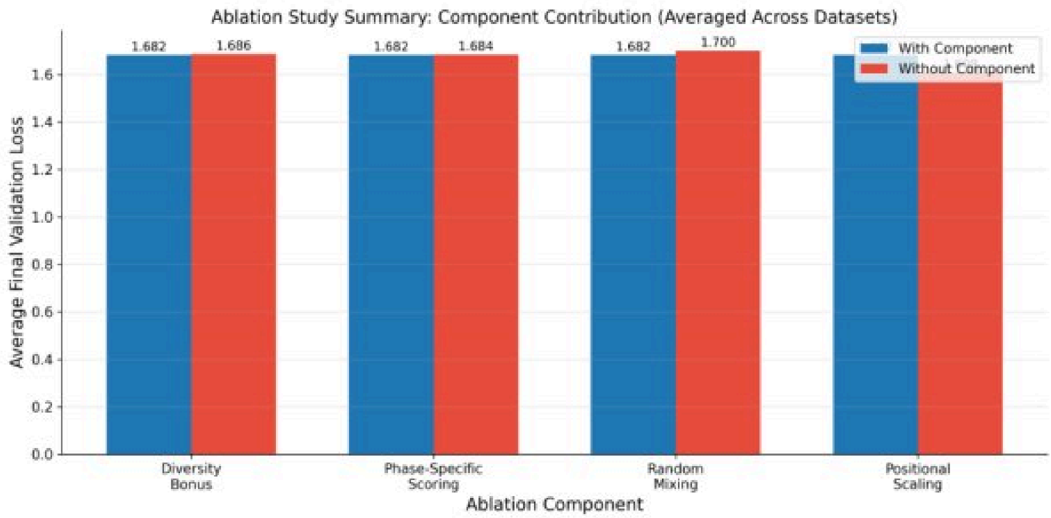

Figure 7: Summary of ablation study effects on average final validation loss.

C Training Efficiency Analysis

Figure 10 shows training efficiency over epochs, with all methods showing similar efficiency decay patterns.

D Implementation Details

Model Architecture. Our Transformer uses learned positional encodings initialized with $\mathcal{N}(0, 0.02)$, GELU activations, and dropout rate 0.1. The embedding dimension is scaled by $\sqrt{d_{\text{model}}}$ following **?**. Total parameters: approximately 2.1M.

Phase Detection Thresholds. We classify epochs as memorization if average improvement > 0.02, generalization if average improvement > 0.005 or improvement standard deviation > 0.01, and refinement otherwise. Window size $w = 5$ epochs provides stable phase detection.

Scoring Functions. In memorization phase: $s_i = \ell_i \cdot g_i \cdot d_i$. In generalization phase: $s_i = -|\ell_i - \text{median}(\ell)| \cdot d_i$. In refinement phase: $s_i = \ell_i \cdot d_i$.

Computational Overhead. PADA adds approximately 15% wall-clock overhead compared to random selection due to gradient norm computation and sample scoring.

Dataset Statistics. WikiText-2: 2.5M characters training, 250K validation. AG News: 1.9M characters training, 190K validation. IMDB: 3.2M characters training, 320K validation. All use vocabulary size 256 (character-level) and sequence length 96.

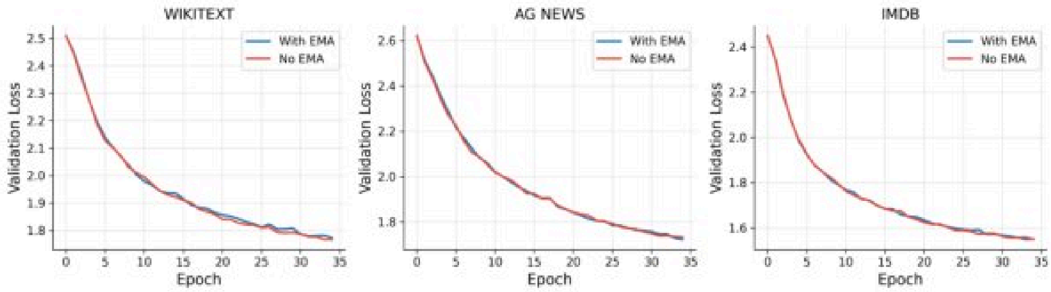

Figure 8: Comparison of sample statistic tracking with and without EMA across all datasets.

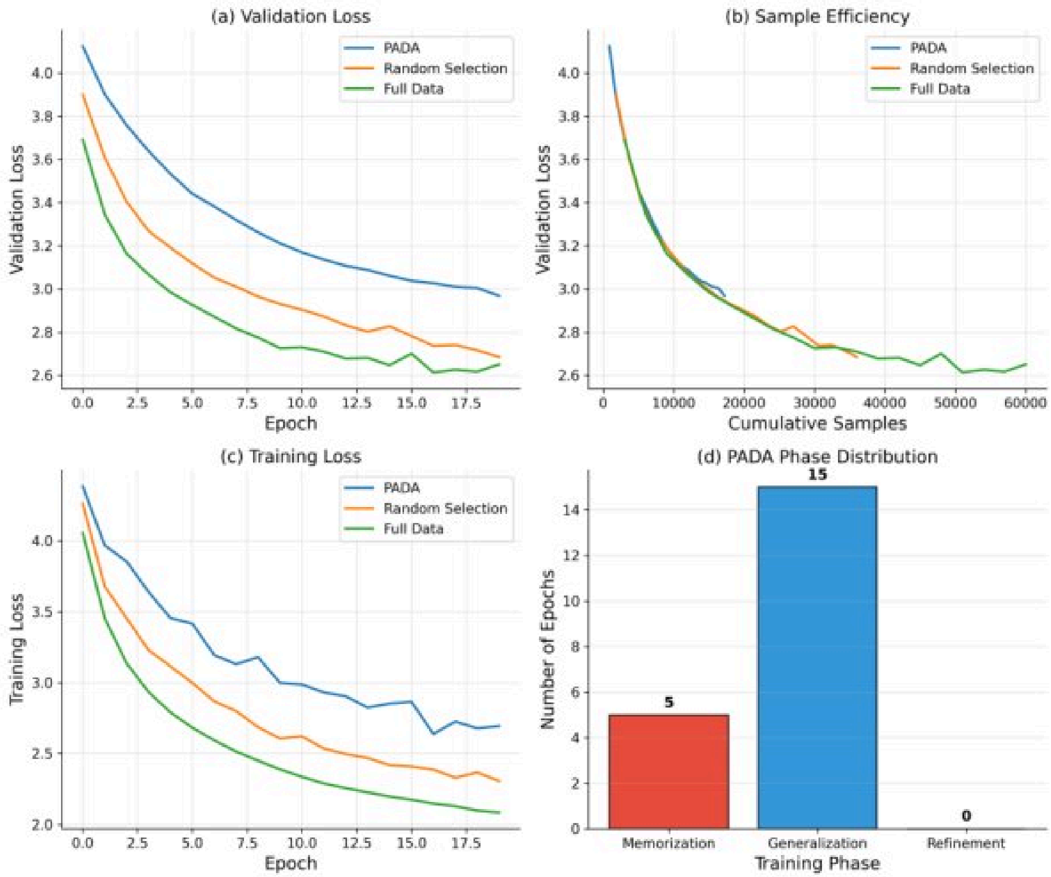

Figure 9: Baseline results on synthetic data showing PADA performs substantially worse than random selection and full data.

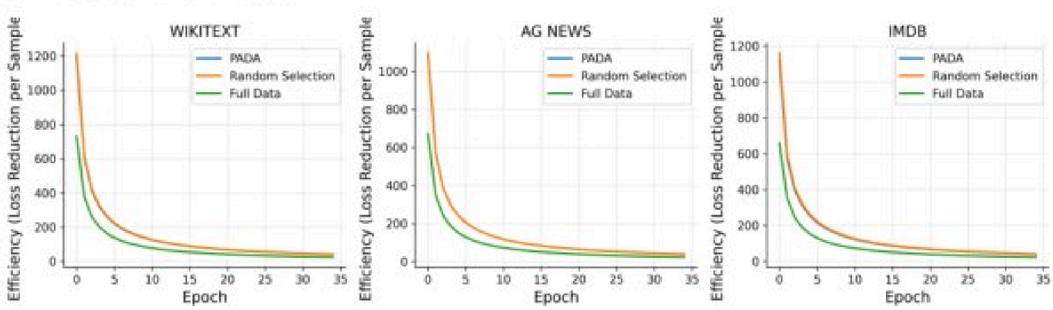

Figure 10: Training efficiency (validation loss reduction per sample) over epochs for all datasets.